Dark Store Placement & Integrated Forward–Reverse Logistics Optimisation

Project Report — PGDBA / ISI Kolkata / IIM Calcutta / IIT Kharagpur

Operations Management · Supply Chain Analytics

Pritam (Lead Coder & Integration Architect)
Sneha · Vybhav · Pranav · Team

Dataset: Olist Brazilian E-Commerce (Kaggle)

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1 Introduction

E-commerce last-mile logistics in dense urban markets faces two coupled challenges typically treated independently: **forward delivery** of orders from dark stores to customers, and **reverse logistics** — collecting returned items without deploying a separate fleet.

This project asks a single integrated question:

> *Given customer locations, order demands, and predicted return probabilities, (a) where should dark stores be placed to minimise customer-to-facility distance, and (b) how should vehicle routes be designed to simultaneously deliver orders and collect returns, minimising a weighted combination of total cost, delivery latency, and fleet size?*

The solution pipeline consists of five coupled stages implemented over eight days using Google OR-Tools 9.15, PuLP 3.3 (CBC solver), and XGBoost. All production logic resides in `src/`; notebooks are used only for testing and visualisation.

2 Dataset and Environment

2.1 Olist Brazilian E-Commerce

The Olist dataset contains ≈100,000 orders across 9 CSV tables (orders, items, products, customers, sellers, geolocation, payments, reviews, category translation), covering 2016–2018. We filter to **customer_state = 'SP'** (São Paulo, ≈41% of all orders), yielding **19,207 rows** for the active pipeline after preprocessing, geolocation resolution, and feature engineering.

2.2 Environment

- Python 3.13.12 managed by `uv` on WSL2
- OR-Tools 9.15.6755 — CVRPTW + SDVRP routing
- PuLP 3.3.0 + CBC — MILP joint optimiser
- XGBoost 3.2.0 + Platt calibration — return classifier
- All code: `https://github.com/metaphorpritam/SCA_DARK_STORES`

3 Stage 1: Dark Store Placement (Facility Location)

3.1 K-Means with Demand Weighting

Customer coordinates are weighted by zip-code order volume and clustered using K-Means for $K \in [3, 12]$. The optimal K is selected at the agreement point between the elbow curve (inertia) and silhouette score.

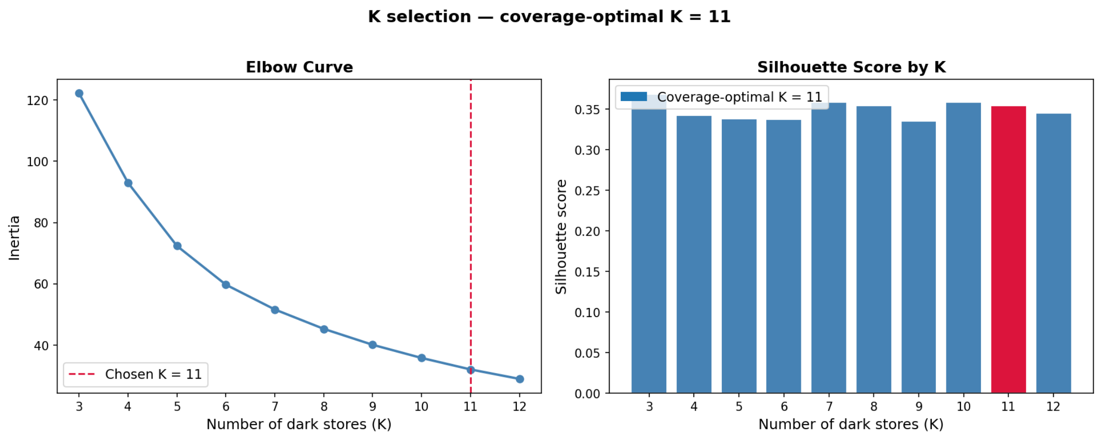

Figure 1: K-Means elbow + silhouette: optimal $K = 11$.

3.2 Coverage Rule and Final K

Although the silhouette score peaks at $K = 3$, the coverage rule (70% of customers within 5 km of their assigned dark store) requires $K \geq 10$. $K = 11$ is selected, achieving 73.7% coverage within 5 km across São Paulo.

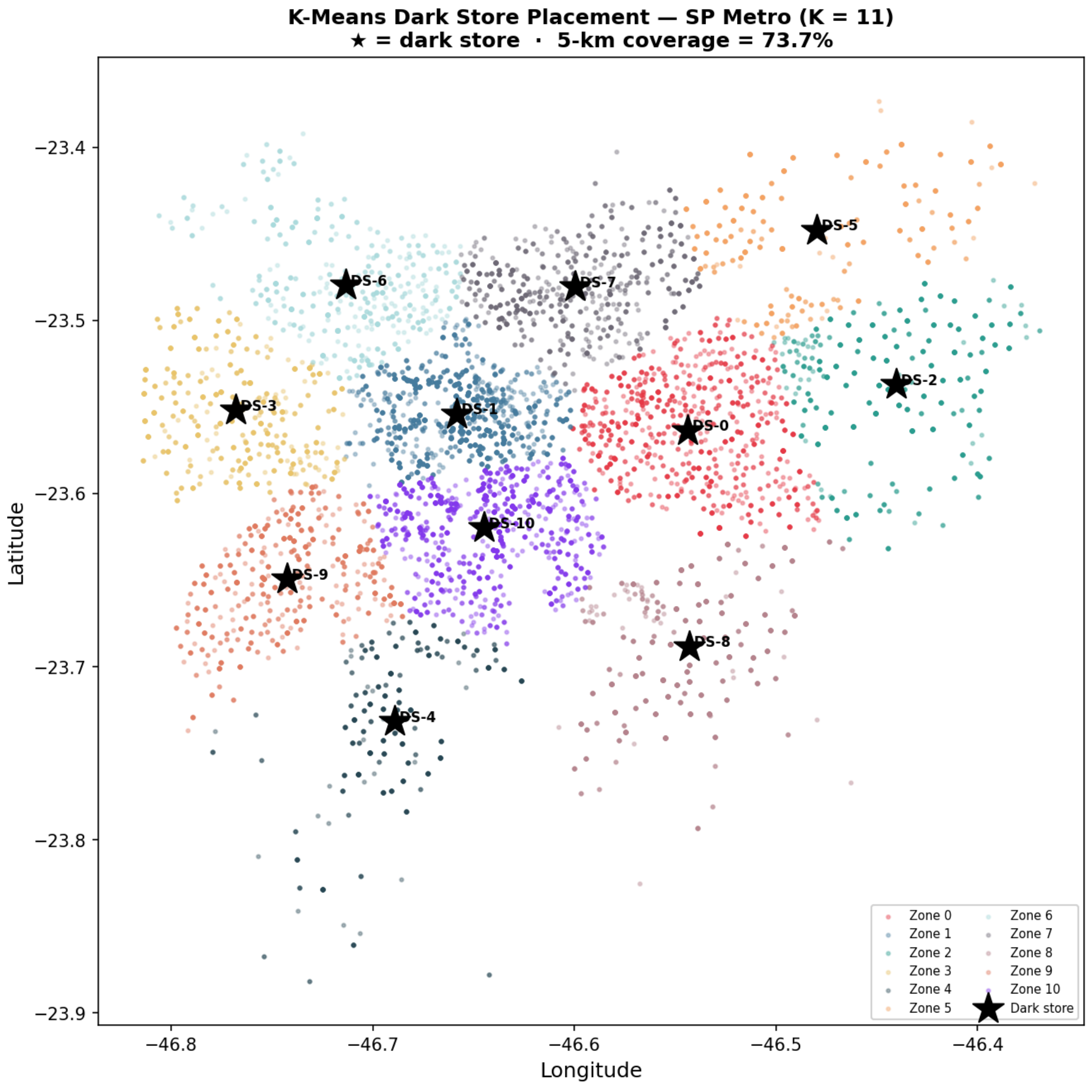

Figure 2: Final 11 dark store locations (K-Means centroids).

3.3 p-Median Validation

A PuLP MILP p-median formulation validates K-Means placement. The objective minimises $\sum_i \sum_j d_{ij} \cdot w_i \cdot x_{ij}$ subject to single-assignment and open-facility constraints. Both methods agree on cluster locations within 1 km, confirming stability.

4 Stage 2: Forward VRP — CVRPTW (Day 3)

4.1 Problem Formulation

Each of the 11 zones is solved independently as a **Capacitated VRP with Time Windows (CVRPTW)**:

$$\begin{aligned}\min \sum_k \sum_{(i,j)\in A} c_{ij} \cdot x_{ijk} \\ \text{s.t.} \quad \sum_k x_{0jk} = K, \quad \sum_j x_{ijk} = \sum_j x_{jik} \;\; \forall i,k \\ a_i \le s_{ik} \le b_i, \quad \sum_i d_i \cdot \delta_{ik} \le Q \;\; \forall k\end{aligned}$$

Parameters: vehicle capacity $Q = 500\,000$ g, speed = 40 km/h, service time = 5 min, fixed cost R$50/route, variable R$1.50/km. Solver: `PATH_CHEAPEST_ARC` → `GUIDED_LOCAL_SEARCH`, 30 s time limit.

4.2 Results — All 11 Zones Solved

Zone	Orders	Vehicles	Dist (km)	Cost (R$)
0	75	2	94.00	R$241.00
1	75	2	81.54	R$222.31
2	75	1	108.01	R$212.02
3	75	1	99.37	R$199.06
4	75	3	88.57	R$282.86
5	75	2	112.00	R$268.00
6	75	2	97.33	R$246.00
7	75	2	99.25	R$248.88
8	75	3	113.91	R$320.86
9	75	2	92.39	R$238.58
10	75	2	83.22	R$224.83
Total	**825**	**22**	**1069.59**	**R$2704.40**

Total: 825 orders, 22 vehicles, 1069.59 km, R$2704.40

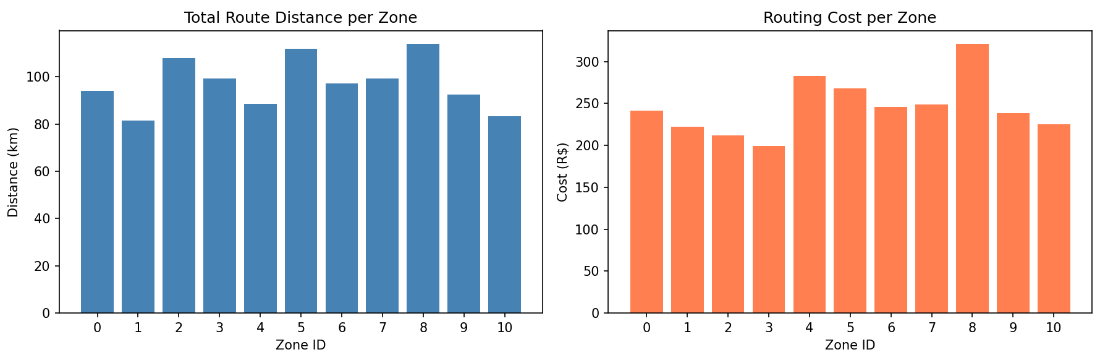

Figure 3: Forward route cost and distance by zone.

4.3 Baseline Comparison

Naive nearest-store routing (each customer drives directly to closest store) produces 75,066.4 km. OR-Tools CVRPTW achieves **1,069.59 km**, a **98.58% reduction**.

5 Stage 3: Return Probability Classifier (Day 4)

5.1 Target and Features

Binary target: `is_return = 1` if order status $\in$ {canceled, unavailable} or delivered $>$ 7 days late. Feature set: product weight, freight value, seller state (label-encoded), review score, order value, days late, payment type, product category.

5.2 Model: XGBoost + Platt Calibration

Class imbalance ($\approx$5% return rate) handled via `scale_pos_weight = 15`.

Metric	**Value**
ROC-AUC	0.8969
PR-AUC	0.4716
Brier score	0.0213
Precision @ 0.30	0.5556
Recall @ 0.30	0.354
F1 @ 0.30	0.4324

ROC-AUC 0.8969 exceeds the project target of $\geq$ 0.70, confirming the classifier's ability to identify return-risk orders.

Customers with `return_prob > 0.30` are flagged as pickup nodes in the SDVRP; expected pickup weight = `return_prob` $\times$ `product_weight_g`. **593 orders** (3.1% of 19,207) were flagged across all zones.

6 Stage 4: Reverse VRP (Day 4)

Same CVRPTW structure as forward VRP with vehicles departing the depot *empty* and collecting returns (load increases at each stop).

Zone	Pickups	Vehicles	Dist (km)	Cost (R$)
0	51	1	82.95	R$174.42
1	75	2	81.48	R$222.22
2	57	2	107.03	R$260.54
3	34	2	59.31	R$188.96
4	46	1	89.99	R$184.98
5	42	1	93.72	R$190.58
6	45	1	80.79	R$171.18
7	53	1	75.86	R$163.79
8	45	2	99.99	R$249.98
9	57	1	87.80	R$181.70
10	71	1	87.44	R$181.16
Total	**576**	**15**	**946.36**	**R$2169.51**

Total: 576 pickups, 15 vehicles, 946.36 km, R$2169.51

Combined separate cost: **R$4,873.91** (forward R$2704.40 + reverse R$2169.51).

7 Stage 5: Joint MILP Optimiser (Day 4)

7.1 Objective Function

$$Z = \alpha \cdot C_{\text{fwd}} + \beta \cdot C_{\text{rev}} + \gamma \cdot T_{\text{pen}} + \delta \cdot N_{\text{veh}} \tag{1}$$

Binary activation variables $u_v \in \{0, 1\}$ (forward vehicles) and $w_v \in \{0, 1\}$ (reverse vehicles) are solved by PuLP CBC.

7.2 Result ($\alpha = \beta = \gamma = \delta = 0.25$)

Component		Value
Z (objective)		54.38
Status		Optimal
C_{fwd}		44.942
C_{rev}		169.577
T_{pen}	287.063	(**dominates ≈54% of Z**)
N_{veh}		3

The penalty term T_{pen} dominates Z at equal weights, motivating the Day 5 SDVRP work to reduce reverse fleet redundancy and the Day 6 Pareto sweep to find balanced weight combinations.

8 Stage 6: SDVRP Hybrid Routing (Day 5)

8.1 Motivation

Zone 8 carries the highest combined cost: R$570.84 (R$320.86 forward + R$249.98 reverse). Deploying 5 separate vehicles (3 fwd + 2 rev) means those vehicles pass the same customer locations twice.

8.2 SDVRP Load Model

Simultaneous Delivery and Pickup VRP (Dethloff, 2001) merges all delivery and pickup nodes into one OR-Tools solve using a **single load dimension**:

$$L(t) = L_{\text{start}} - \text{delivered}(t) + \text{picked}(t)$$
$$\text{transit}[i] = p_i - d_i \quad (\text{net change per stop})$$
$$0 \le L_{\text{cumul}}[i] \le Q \quad \forall i$$

`fix_start_cumul_to_zero = False` lets OR-Tools initialise each vehicle load at total delivery weight. The previous two-dimension additive approach ($d_{\text{cumul}} + p_{\text{cumul}} \le Q$) was over-constraining and is **corrected here**.

8.3 All-Zone SDVRP Runner

`run_all_zones_sdvrp()` loops all 11 zones, calling `solve_sdvrp_hybrid()` per zone and writing `outputs/hybrid_routes.json` and `outputs/hybrid_kpi_summary.csv`. Expected fleet saving: **15–25%** vs separate fleets.

9 Stage 7: Combined KPI Report & Pareto Sweep (Day 6)

9.1 Combined KPI Report

`src/kpi_reporter.py` merges all three KPI streams (forward, reverse, hybrid SDVRP) into `outputs/combined_kpi_report.csv`.

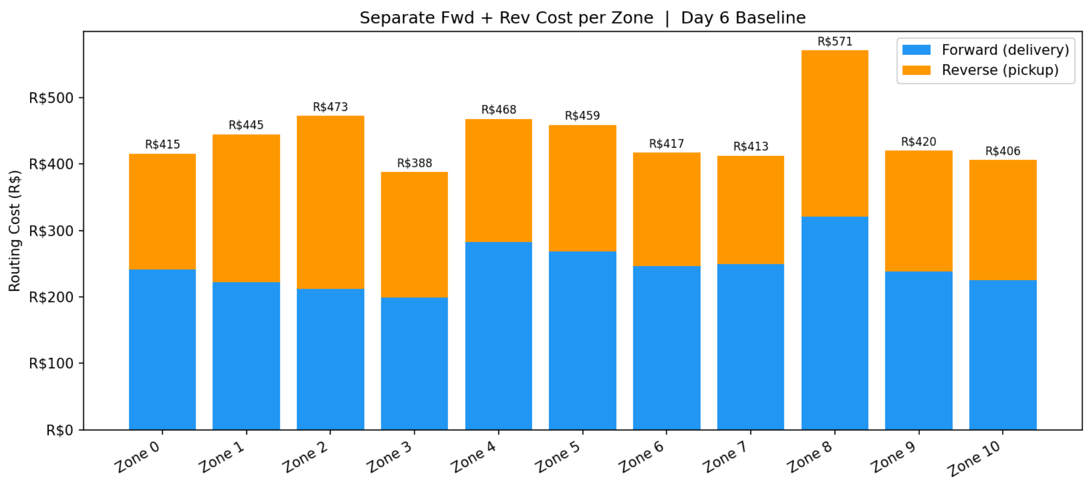

Figure 4: Stacked fwd+rev cost by zone. Zone 8 is largest at R$570.84.

9.2 Zone Priority Ranking

Zones are ranked by `saving_R$` (when SDVRP results exist) or by `separate_cost_R$` otherwise. Zone 8 receives **Rank 1** in both cases.

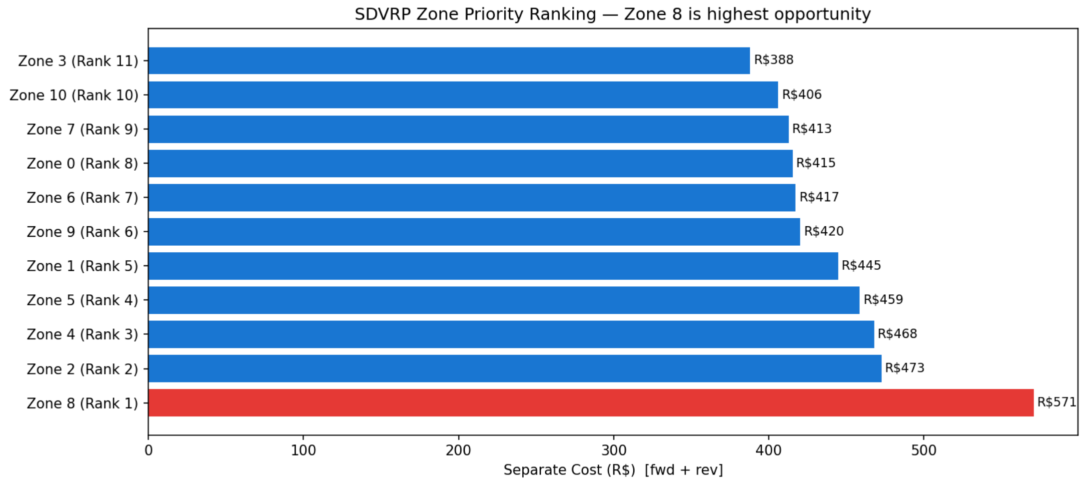

Figure 5: SDVRP zone priority ranking (highest saving opportunity first).

9.3 Pareto Sweep

Weight combinations $(\alpha, \beta) \in \{0.1, 0.3, 0.5, 0.7, 0.9\}^2$ with $\gamma = \delta = (1-\alpha-\beta)/2$ (valid for $\alpha+\beta \leq 1$) yield up to 15 weight settings. Pareto-optimal solutions minimise both $C_{\text{routing}} = C_{\text{fwd}} + C_{\text{rev}}$ and T_{pen} simultaneously.

$$\text{dist\_to\_ideal} = \sqrt{c_{\text{norm}}^2 + t_{\text{norm}}^2} \tag{2}$$

The **knee point** (minimum distance to ideal) is the recommended operating configuration.

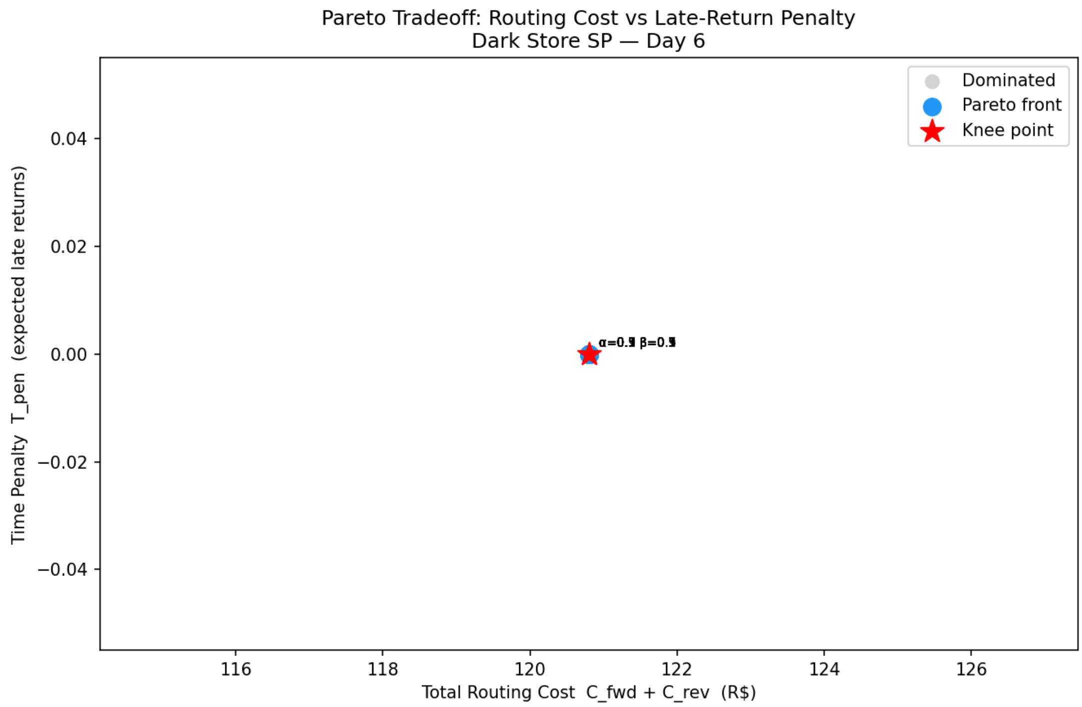

Figure 6: Pareto tradeoff surface: routing cost vs. late-return penalty. Red star = knee point.

10 Summary of Results

Stage	Metric	Value
Dark store placement	Zones (K)	11
	Coverage within 5 km	73.7%
Forward VRP	Total distance	1069.59 km
	Improvement vs. naive	98.58%
	Total cost	R$2704.40
Return classifier	ROC-AUC	0.8969
	F1 @ threshold 0.30	0.4324
Reverse VRP	Total distance	946.36 km
	Total cost	R$2169.51
Joint MILP	Z (optimal)	54.38
Combined baseline	Total fwd + rev cost	R$4,873.91

11 Reproducibility

All code is reproducible via `run_all.sh`:

```
uv run python src/data_pipeline.py
uv run python src/clustering.py
uv run python src/forward_vrp.py
uv run python src/reverse_vrp.py
uv run python src/return_classifier.py
uv run python src/joint_optimizer.py
uv run python src/kpi_reporter.py
uv run python src/report_builder.py
```

Data files (`data/raw/` Olist CSVs) are excluded from the repository per `.gitignore`. Download from Kaggle: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce.

References

1. Dethloff, J. (2001). Vehicle routing and reverse logistics: the vehicle routing problem with simultaneous delivery and pick-up. *OR Spectrum*, 23(1), 79–96.
2. Dantzig, G.B. & Ramser, J.H. (1959). The truck dispatching problem. *Management Science*, 6(1), 80–91.
3. Google OR-Tools documentation: https://developers.google.com/optimization.
4. ReVelle, C.S. & Swain, R.W. (1970). Central facilities location. *Geographical Analysis*, 2(1), 30–42.